

LMCP AUTOQUOTE EXECUTIVE SYSTEM BLUEPRINT

Autonomous South African supply-and-delivery tender intelligence, quotation production, and submission operating blueprint

EXECUTIVE SUMMARY

Purpose-built to harvest qualifying public opportunities, apply commercial decision rules, generate branded quote packs, and prepare submission-ready email workflows with traceable control.

PREPARED FOR

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COMPANY

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DATE

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PURPOSE

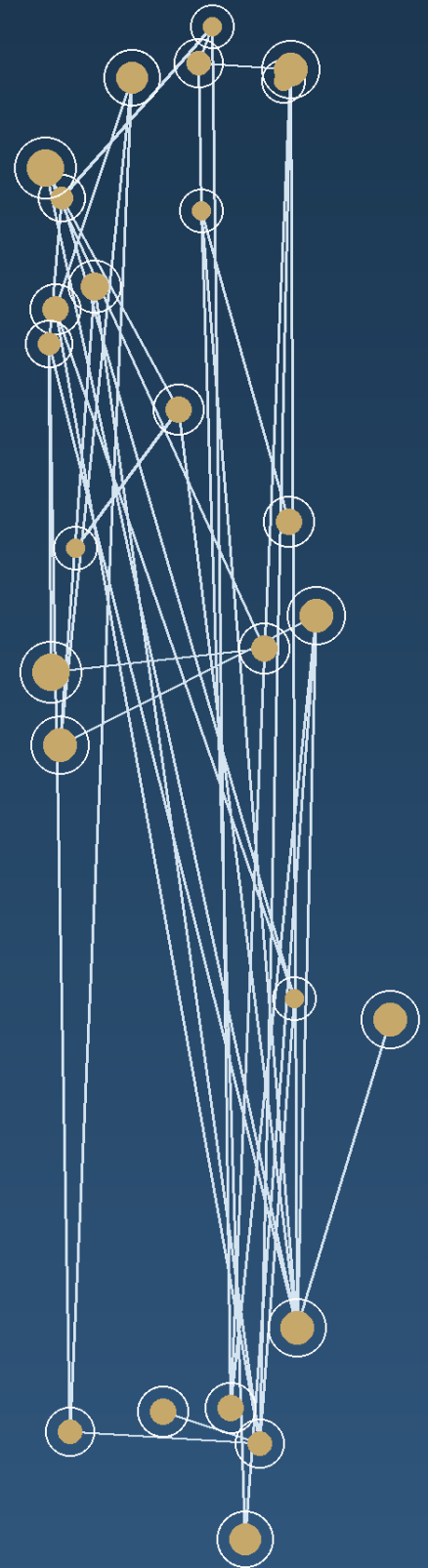
Final build, integration, deployment, and controlled scale-up

OPERATING FOCUS

Profitable, email-submittable public supply-and-delivery work in South Africa

Executive architecture document

Prepared from the full LMCP AutoQuote system blueprint



1. Executive brief

Operating mandate, commercial intent, and scope boundaries for the canonical production path.

LMCP AutoQuote is envisioned as a focused tender operations engine for profitable public supply-and-delivery work in South Africa. The platform should continuously harvest opportunities, apply explicit qualification rules, estimate commercial viability, generate branded quotation packs, and prepare submission-ready email workflows inside one controlled operating model.

This is not a generic tender scraper. It is a production system for finding the right opportunities, rejecting the wrong ones early, and moving only qualified work into pricing, pack generation, and dispatch with full auditability.

OPERATING MANDATE

One canonical workflow should serve manual run-once execution, scheduled autonomous harvesting, and supervised auto-submission. All three operating modes should use the same services, data objects, and decision rules.

Business goal

- Continuously discover qualifying RFQs and tender notices from approved public-sector procurement sources.
- Advance only opportunities that match LMCP target categories and email-submittable operating constraints.
- Turn approved opportunities into branded PDF quote packs with traceable submission preparation.

Success criteria

- Clear rule-based qualification with explainable exclusion reasons.
- Reliable commercial scoring and quote-readiness promotion.
- Controlled pack assembly, attachment manifesting, and email dispatch logging.

2. System at a glance

End-to-end operating chain and scope guardrails.

End-to-end operating pipeline

One canonical workflow should power manual runs, scheduled harvesting, and supervised auto-submission.



Control principle All entry points - dashboard actions, API endpoints, Celery schedules, and supervised auto-runs - should call the same orchestration layer so that qualification, pricing, pack assembly, and submission rules remain consistent.

In scope

- Supply and delivery RFQs and tenders across national, provincial, municipal, SOE, and other approved public-sector sources.
- Harvesting, normalization, qualification, relevance and profitability screening, quote drafting, PDF generation, compliance assembly, and submission readiness.
- Reusable company branding, compliance document storage, and approved supplier or product pricing logic.

Out of scope for current production track

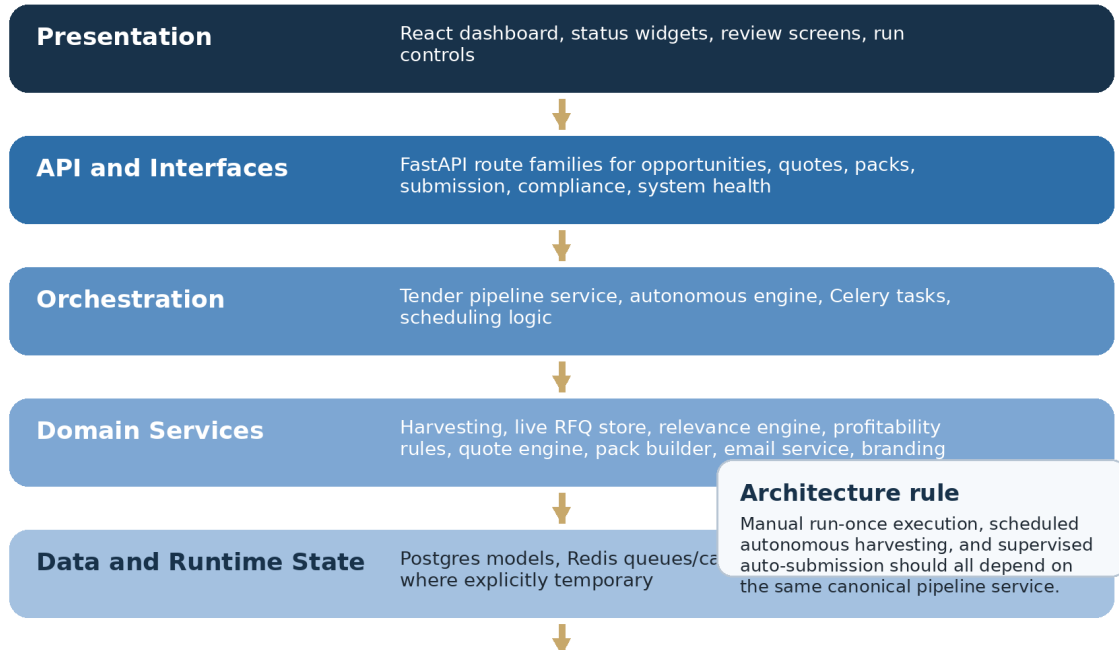
- Medical consumables, IT equipment, petrol, diesel, and other excluded categories.
- Any opportunity with a compulsory briefing session or site visit requirement.
- Portal-native workflows that cannot be translated into a controlled email submission path.

3. Target architecture

Single-stack design to prevent duplication, preserve consistency, and support controlled autonomy.

Target architecture

A single service stack with clear boundaries prevents duplicated logic and preserves auditable workflow control.



Core layer map

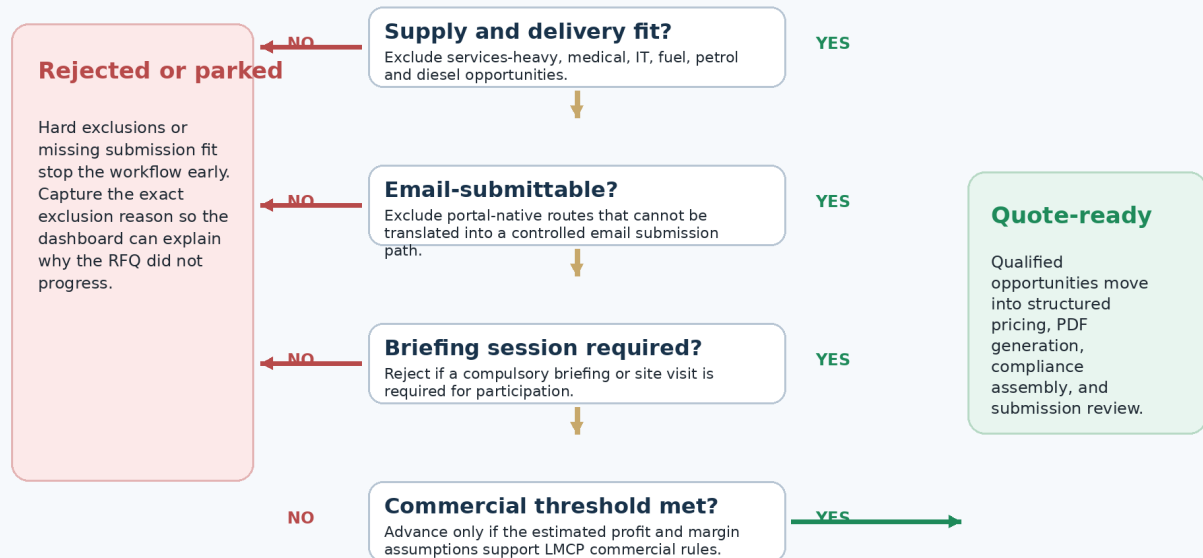
Layer	Purpose	Primary components
Presentation	Human control and visibility	React dashboard, status widgets, review screens, run controls
API and interfaces	Secure business entry points	FastAPI route families for dashboard, opportunities, quotes, packs, submission, compliance, system health
Orchestration	Cross-module workflow execution	Tender pipeline service, autonomous engine, Celery tasks, scheduling logic
Domain services	Business logic and transformations	Harvester, live RFQ store, relevance engine, profitability rules, quote engine, pack builder, email service, branding
Data and runtime state	Persistence and caching	Postgres models, Redis queues and cache, tightly controlled temporary runtime JSON
External integrations	Connectivity to sources and delivery channels	eTenders adapters, other approved portal adapters, SMTP provider, document generation libraries

4. Commercial decisioning

Qualification, exclusion, and quote-readiness promotion should be rule-based and explainable.

Qualification and decision logic

Hard exclusions fire first. Commercial scoring only begins once an opportunity fits the target operating model.



Decision framework

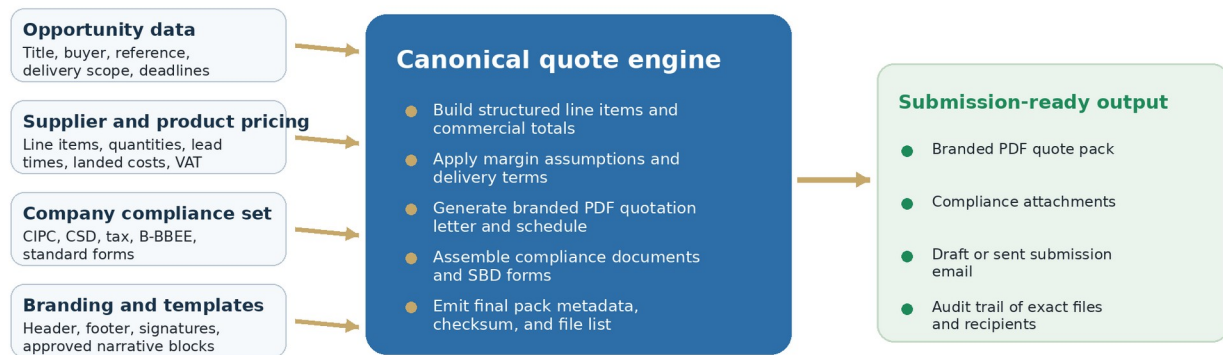
Control area	Rule intent	Operational consequence
Hard exclusions	Reject services-heavy work, excluded categories, briefing-session tenders, and non-email-compatible routes.	Stops wasted effort before scoring and stores the exact exclusion reason.
Positive signals	Prefer goods-based procurement, repeatable delivery scope, supplier-aligned categories, and submission routes that can be handled reliably.	Raises relevance and reduces manual interpretation burden.
Commercial threshold	Advance only when estimated profit and margin assumptions meet LMCP commercial rules.	Promotes work into quote-ready state and avoids low-value churn.
Explainability	Every decision outcome must be traceable in both machine-readable and human-readable form.	Supports dashboard review, audits, debugging, and operating trust.

5. Quote generation and submission pack

The quote factory must be data-driven, brand-safe, and compliance-aware.

Quote pack factory

The quote pack must be produced from real line-item data, approved compliance documents, and consistent branding control.



Quote PDF standard

- LMCP-branded header and identity pulled from a central branding module.
- Opportunity reference, issuing entity, RFQ number, and closing information.
- Structured line items, unit pricing, delivery assumptions, VAT, totals, and validity period.
- Authorised signature block and approved narrative language.

Submission pack standard

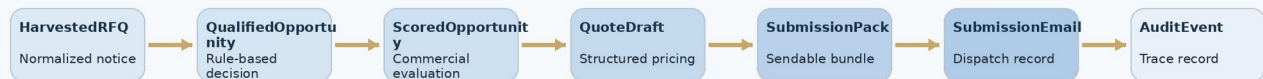
- Primary quotation PDF plus the commercial offer.
- Required compliance attachments and completed standard forms where applicable.
- Exact attachment manifest with final filenames, checksum, and storage path.
- Email draft or sent record linked to the same bundle metadata.

6. Core data objects and audit trail

Objects should progress through a controlled lifecycle with persistent references to files and outcomes.

Core data objects and audit trail

Objects progress through a controlled lifecycle so every decision, file, and email can be explained later.



Audit principle Every transition - qualification, score change, quote generation, pack assembly, submission, attachment manifest, and delivery result - should create an audit event linked back to the opportunity and the files that were actually sent.

Canonical data objects

Object	Purpose	Key fields	Lifecycle notes
HarvestedRFQ	Normalized notice record	external_id, source, title, issuing_entity, province, dates, submission_method	Created by the harvester; raw source reference retained.
QualifiedOpportunity	RFQ after rule-based qualification	qualification_status, exclusion_reason, commodity_tags, email_submission_ok	Transitions from harvested state after rule evaluation.
ScoredOpportunity	Commercially evaluated opportunity	relevance_score, profit_estimate, margin_assumption, quote_ready	Drives queue promotion into the quote stage.
QuoteDraft	Structured quotation dataset	items, quantities, unit prices, totals, VAT, delivery assumptions, validity	Can be regenerated after pricing changes.
SubmissionPack	Final sendable bundle metadata	quote_pdf_path, compliance_files, sbd_forms, checksum, status	Must track the exact files sent.
SubmissionEmail	Email dispatch record	recipient, cc, subject, attachment_paths, sent_at, result	Should support both draft and sent states.
AuditEvent	Trace log	event_type, entity_type, entity_id, timestamp, actor, metadata	Generated at each significant stage.

7. Canonical modules and route families

Responsibilities should stay singular and interfaces should stay predictable.

Module responsibility map

Module or area	Canonical responsibility
app/main.py	Single application entry point with approved routers and startup hooks.
app/tasks.py	Celery task registration and task-level orchestration entry points.
app/services/tender_pipeline.py	Canonical orchestrator from harvested items to qualified, quote-ready output.
app/services/tender_harvester.py	Harvests and normalizes notices from source adapters.
app/services/live_rfq_store.py	Temporary live RFQ runtime store and filtered retrieval helpers.
app/autonomous_engine.py	Scheduling control and supervised autonomous pipeline execution.
app/quote_pack_service.py	Document assembly, pack generation, and final output handling.
app/email service or email_api.py	Submission drafting, sending, and message logging.
app/branding/company_branding.py	Central identity and template configuration used by all outgoing documents.

API route blueprint

Route family	Purpose	Control note
dashboard	Summary, health, counts, latest events	Read-only operational visibility.
autonomous	Status, run-once, schedule control	Must call the canonical pipeline service.
opportunities	Harvest results, review, promotion	Supports browsing and triage without bypassing rules.
quotes	Quote draft generation and retrieval	Should not skip qualification data.
quote pack	Build and retrieve PDF packs	Uses real quote data, branding, and manifests.
submission	Draft or send emails; track results	Must create audit events and retain file references.
compliance	Manage compliance documents and requirement mapping	Central source for attachment assembly.
system	Readiness and diagnostics	Used for environment and service checks.

8. Governance, security, and control tower

Production-safe autonomy depends on explicit safeguards, auditability, and human review points.

Security and governance controls

- Protect email credentials, database secrets, and external keys with environment-bound secret handling.
- Validate outbound recipient addresses against approved instructions before sending.
- Preserve source provenance and file lineage so every attachment in a pack can be traced to a stored reference.
- Never allow silent replacement of quote tools, recipients, or attachments without an audit event.

Control tower requirements

- System health tiles for API, database, Redis, worker, and scheduler readiness.
- Batch counters for harvested, excluded, qualified, quote-ready, quoted, packed, emailed, and failed states.
- Latest error stream with drill-down to entity and stage.
- Review queue for high-risk steps, retries, and approval gates before auto-submission.

DESIGN PRINCIPLES

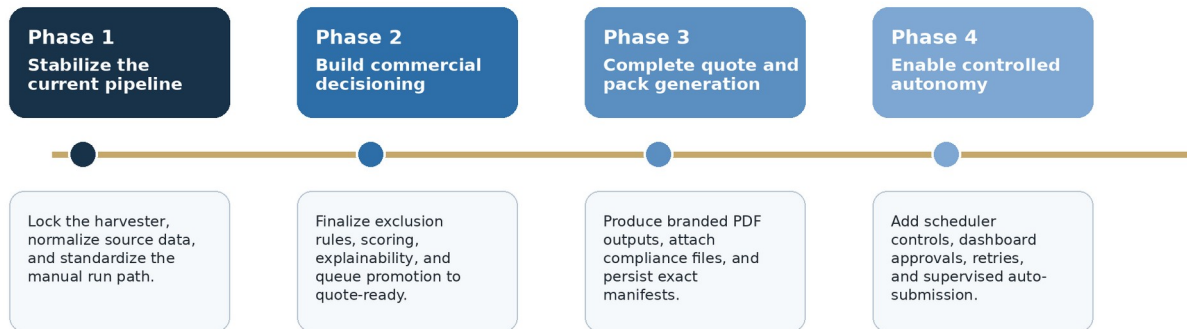
Single source of truth. No duplicate systems. Deterministic pipeline behavior. Traceability first. Production-safe autonomy. Brand and compliance integrity.

9. Recommended build sequence and acceptance

Finish the canonical chain first, then add scale and autonomy on top of a stable operating path.

Recommended build sequence

Finish the canonical chain first. Scale only after qualification, quote generation, and submission controls are reliable.



Non-negotiable acceptance criteria

- A single manual trigger can execute the pipeline from harvest to quote-ready without code edits.
- Qualified opportunities are explainable and visibly filtered against LMCP rules.
- Generated quotation PDFs contain real data and correct totals.
- Submission emails can be drafted or sent with the exact quote PDF and required compliance files attached.

Future expansion path

- Broaden source coverage to additional SOEs, departments, and municipalities once the core pipeline is reliable.
- Add richer supplier catalogs and partial pricing intelligence.
- Introduce approval workflows for higher-risk submissions.
- Expand reporting and dashboards once the canonical production path is stable.

10. Final implementation note

The blueprint prioritizes disciplined consolidation over parallel experimentation.

The fastest path to a reliable LMCP AutoQuote production system is disciplined consolidation: qualification, scoring, quote drafting, branded PDF generation, compliance assembly, and submission logging should all remain inside one canonical flow.

If a proposed change does not clearly strengthen that chain, reduce risk inside that chain, or improve observability around that chain, it should be deferred until the core path is stable.

EXECUTIVE CONCLUSION

Build one operating machine: harvest -> qualify -> score -> quote -> brand -> attach -> submit -> audit. Make it visible, explainable, and commercially disciplined before adding scale.

Immediate next actions

Priority	Action	Outcome
1	Stabilize harvester inputs and normalized RFQ schema	Consistent source data and reliable manual run path
2	Finalize exclusion rules, scoring, and explainability	Controlled opportunity promotion to quote-ready
3	Complete branded quote pack generation and attachment manifesting	Submission-ready document output with traceable files
4	Enable scheduler controls, dashboard review, and supervised auto-submission	Safe operational autonomy with approval points